

Attractor Representation of Manifolds in Neural Circuits

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Abstract Continuous Attractor Networks (CAN) have proven to be productive in neuroscience in explaining neural dynamical systems that maintain representations of variables in state spaces. In this dissertation project, we aim to extend the CAN framework to investigate the dynamics of neural circuits that represent state spaces that are non-Euclidean manifolds. We first establish a method to train attractor networks to represent such manifolds with minimal assumptions (Michaelmas), then investigate how the brain learns these representations in the motor control context (Lent).

1 Introduction

The brain is often thought to construct internal representations for variables in the external world, in order to perform efficient cognitive/motor control [1]. These variables potentially live in different state spaces, many of which are non-Euclidean manifolds, such as the space of orientations (1D ring) or the space of 3D rotations ($SO(3)$). If/how populations of neurons represent these manifolds is a fundamental building block of understanding how some brain area is involved in particular cognitive/motor functions. To answer these questions, the Continuous Attractor Network (CAN) framework is highly productive in making testable predictions about neural systems that maintain representation of continuous variables in state spaces [2].

Namely, the CAN framework considers the neural circuit as a dynamical system and places two postulates: the state space can be mapped to a (continuous) set of stable equilibria in the neural dynamics, called the attractor manifold, and actions on the external variable correspond to movements along the attractor manifold. One can hand-design neural networks that satisfy these postulates, as is done in classical exposition of the ring attractor model [3].

However, this type of hand-tuning can be cumbersome and may not generalize well to more complex manifolds. Instead, one may consider optimizing the neural network to represent the manifold directly, by training the network on some objective function that ensures satisfaction of the postulates. This has been done previously, but most existing methods supervise the network to explicitly predict positions on the manifold given sequences of actions. This imposes inductive biases on the network up to the parametrization of the manifold, which may not be desirable when no/multiple parametrizations are possible. For example, $SO(3)$ can be parametrized by Euler angles, quaternions, or rotation matrices, and each parametrization has its own geometry.

In the first half of this project, we establish a method based on contrastive learning to train neural networks to represent manifolds with minimal constraints. This is pursued in Michaelmas term and is described in Sections 2 and 3, representing the bulk of this report. The next step is to investigate how the brain learns these representations in the motor control context, which is pursued in Lent term and discussed in Section 4.

2 Methods

In this section and onwards, we will use the following notation conventions. Positions on the manifolds are denoted by $x \in \mathcal{M}$, where $\mathcal{M}$ is the manifold. Actions on the manifold are denoted by $u \in \mathcal{U}$, where $\mathcal{U}$ is some parametrization of the action space (for

the manifolds explored thus far, $\mathcal{U}_x$ is the Euclidean tangent space of $\mathcal{M}$ at position x). The neural representation is denoted by $g \in \mathbb{R}^d$, where d is the dimension of the neural representation. Sequences from time t to $t + T$ are denoted by $\cdot_{t:t+T} = (\cdot_t, \cdot_{t+1}, \dots, \cdot_{t+T})$. We denote the dynamical update function of the neural network as $\mathcal{F}$, such that $g_{t+1} = \mathcal{F}(g_t, \nabla x_t) = \mathcal{F}(g_t, u_t)$.

With this setup, the CAN postulates formalizes as follows.

$$\text{Stability:} \quad \mathcal{F}(g_t, 0) = 0 \quad (1)$$

$$\text{Path Integration:} \quad \int_t^{T_1} u_\tau d\tau = \int_t^{T_2} v_\tau d\tau \iff \mathcal{F}(g_t, \int_t^{T_1} u_\tau d\tau) = \mathcal{F}(g_t, \int_t^{T_2} v_\tau d\tau) \quad (2)$$

where integration is performed on the manifold, such as by consulting its Lie algebra.

2.1 Training

We aim to train a neural network to respect Equations (1) and (2) without direct supervision on manifold positions. Instead, the network is trained to integrate actions on the manifold as appropriately as possible. Trajectories of actions are sampled by an AR(1) process, such that $u_{t+1} = \alpha u_t + \epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \Sigma_{\text{AR}})$ is a Gaussian noise¹. Zero-padding is performed at the end of each trajectory to allow for variable-length sequences.

With this base dataset of trajectories, each gradient step is expressed as following.

$$\Theta_{e+1} = \mathcal{D}(\Theta_e, \mathbf{A}_e, \mathbf{P}_e) \quad (3)$$

Where Θ is the set of parameters of the neural network, $\mathcal{D}$ is the gradient descent function, and $\mathbf{A}, \mathbf{P}$ are the anchor and positive inputs sampled at each epoch, such that $\mathbf{A} = \{u_{1:T}^{(i)}\}_{i \in 1:B}$ and $\mathbf{P} = \{v_{1:T}^{(i)}\}_{i \in 1:B}$, where B is the batch size and $\|\int_1^T u_\tau^{(i)} d\tau - \int_1^T v_\tau^{(i)} d\tau\| < \sigma_x \quad \forall i$, where σ_x is a hyperparameter specifying a threshold distance². Feeding the network with $\mathbf{A}$, the network output is denoted as $\mathbf{G}_\mathbf{A} = \{g_{1:T}^{(i)}\}_{i \in 1:B}$, and similarly for $\mathbf{P}$.

2.2 Loss Function

For two trajectories of actions $u_{t:t+T}, v_{t:t+T}$ on a continuous attractor manifold, Equation (2) implies the relationship in Equation (4) in the limit of small displacement σ_x .

$$\left\| \int_t^T u_\tau d\tau - \int_t^T v_\tau d\tau \right\| < \sigma_x \implies \mathcal{F}(g_t, u_{t:t+T}) \approx \mathcal{F}(g_t, v_{t:t+T}) \quad (4)$$

To ensure the bi-directionality of the relationship, the reverse relationship is also considered.

$$\int_t^T u_\tau d\tau \neq \int_t^T v_\tau d\tau \implies \mathcal{F}(g_t, u_{t:t+T}) \neq \mathcal{F}(g_t, v_{t:t+T}) \quad (5)$$

Based on these postulates, a contrastive loss function is designed to train a neural network to represent and integrate on the manifold. Let $\phi(\cdot, \cdot) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a similarity kernel, such as the cosine similarity or negative Euclidean distance. First, we sample an evaluation time index t^* for each trajectory in the zero-padded region of the

¹In this report, $\alpha = 0.8, \Sigma_{\text{AR}} = 0.2I$ is used. At this stage of the project, these parameters should allow for the input dataset to sample the manifold as uniform as possible, though extensions may be taken for representations where the manifold is sampled with bias.

²This could be set explicitly, up to the chosen manifold parametrization, or as a percentile across the dataset

action sequence and denote the evaluated neural representation as $\mathbf{g}^* = \{g_{t^*}^{(i)}\}_{i \in 1:B}$. The loss function is then defined as in Equation (6).

$$\mathcal{L}_{\text{contrastive}} = \frac{\lambda}{\beta} \sum_{i=1}^B \phi(g_{t^*}^{\mathbf{A},(i)}, g_{t^*}^{\mathbf{P},(i)}) - \log \sum_{k=1}^B \exp\left(\frac{\phi(g_{t^*}^{\mathbf{A},(i)}, g_{t^*}^{\mathbf{P},(k)})}{\beta}\right) \quad (6)$$

Respectively, the first term encourages similarity for representations of similar trajectories, and the second term encourages overall separation for representations.

2.3 Network Architecture

To correctly integrate actions $\{u_t\}_{1:T}$ to update the state representation, the neural circuit must be able to represent not only the present manifold element x_t but also some parametrization of $\mathcal{M}$'s tangent space at x_t . In biology, this dynamical, state-dependent action representation is accomplished by neurons jointly tuned to position and action. For example, in the well-studied ring attractor system in the fly brain, where cells in the ellipsoid body (EB) are thought to dynamically maintain an activity bump from which the fly's current orientation can be decoded, cells on the left/right protocerebral bridge (PB) are jointly modulated by clockwise/counterclockwise (CW/CCW) velocities [4].

Based on this intuition, this project considers a partitioned Recurrent Neural Network (RNN) architecture, where the activity of a subnetwork is multiplicatively modulated by the action at each time step, similar PB cells. This is extended to high-dimensional action spaces by introducing different subnetworks independently modulated by different action dimensions. Mathematically, the network update function is expressed as in Equation (7). Let g^{rep} denote the subnetwork independent from the action, and $g^{\text{in},j}$ denote the subnetwork modulated by the action in dimension $j \in [1, J]$. α is a non-linearity, such as ReLU or tanh.

$$\begin{aligned} g_{t+1} &= \begin{bmatrix} g_{t+1}^{\text{rep}} & g_{t+1}^{\text{in},1} & \dots & g_{t+1}^{\text{in},J} \end{bmatrix}^\top, \quad \text{where} \\ g_{t+1}^{\text{rep}} &= \alpha(W_{rr}g_t^{\text{rep}} + \sum_{j \in [1,J]} u_t^j W_{ir}^j g_t^{\text{in},j} + b_r) \\ g_{t+1}^{\text{in},j} &= \alpha(W_{ri}^j g_t^{\text{rep}} + \sum_{j \in [1,J]} u_t^j W_{ii}^j g_t^{\text{in},j} + b_i^j) \end{aligned} \quad (7)$$

This partitioned setup differs from the more flexible setup used in [5] for a similar path integration problem but where the goal is to study the formation of grid cells. A more flexible update equation allows for more complex modulation by actions. However, under the present focus on discovering minimalistic neural circuits, the simpler setup allows more straightforward predictions and interpretations about biological implementations ³.

3 Results

The network has been trained to represent a number of manifolds, including the 1D ring, 2D torus, 2D heading (S^2), and 3D rotations ($SO(3)$). Principal Component Analysis (PCA) is used to visualize the structure of neural representations corresponding to 10000 random input trajectories whose final positions span each manifold. As shown in Figure 1a, the manifold structure of the neural representation is consistent with the trained manifold. One can check that the intrinsic dimensionality of the neural representations

³In those papers, they set $W(u_t) \triangleq \text{MLP}(u_t)$. The MLP architecture allows for (nonlinear) action modulation of single synapses, which lacks biological credibility.

to be same as the manifolds they represent. The color of each point corresponds to the final position on the manifold, and one can see that the neural representation is able to correctly integrate and distinguish between different positions on the manifold, respecting the periodic topologies.

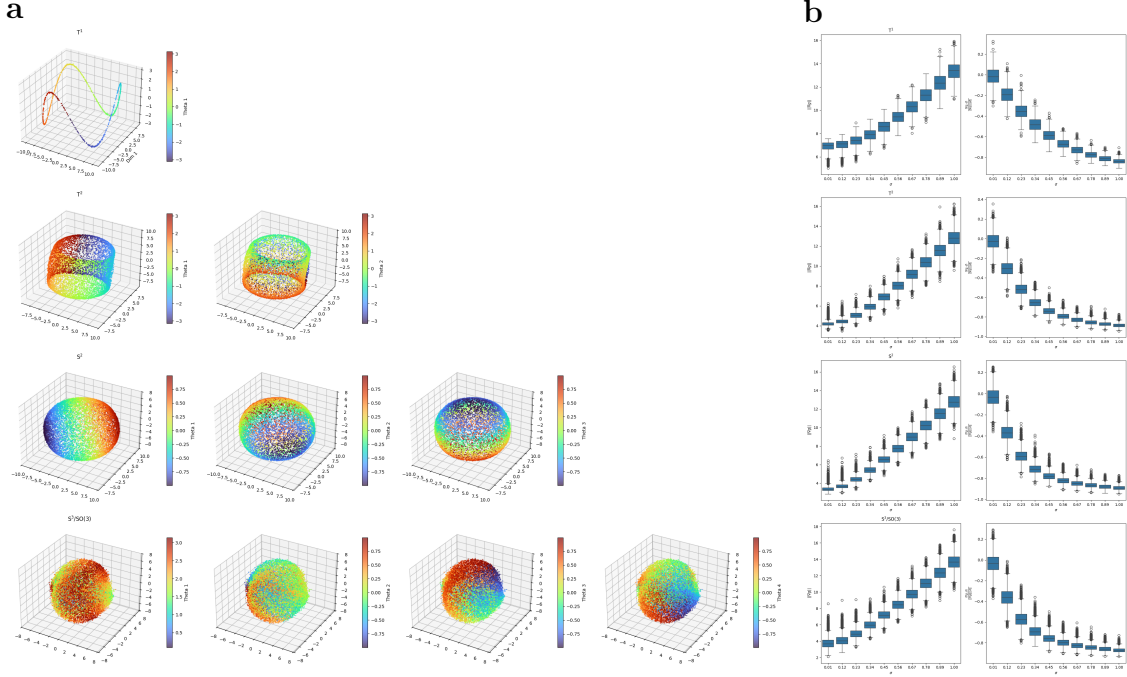


Figure 1 **a** PCA visualization of the neural representation for different manifolds. Each point corresponds to the network’s final activity corresponding to a random input trajectory and is colored by the final position on the manifold. Top to bottom: 1D ring, 2D torus, 2D heading (S^2), and 3D rotations ($SO(3)$, S^3). For $SO(3)$, the position is parametrized by axis-angle representation where "Theta1" is the absolute magnitude of the rotation and "Theta[2, 3, 4]" are the projection of the rotation axis on the x, y, z axes. **b** Gradient against perturbation level for each manifold. Same row arrangement for manifolds as **a**. Left: gradient magnitude against perturbation level. Right: correspondence between the direction of the gradient update and the direction of the perturbation.

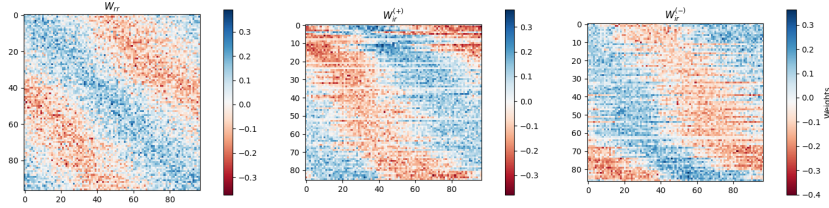
One could also check for the stability of the neural representation by adding different levels of random perturbations $d \sim \mathcal{N}(0, \sigma^2)$ to some neural state on the attractor manifold and observe the direction of the gradient update. In Figure 1b, one notices that the gradient magnitude (left) increases with the average amount of perturbation σ , as the normalized gradient direction (noise) is consistently negatively correlated with the perturbation direction, both signatures of the attractor corresponding to a set of stable equilibria. The additional observation that the alignment is higher for higher perturbation levels is due to that these perturbations are not necessarily orthogonal to the manifold and the attractor dynamics may thus push the representation back to another point on the manifold - this effect is reduced for larger perturbations.

3.1 Comparisons with known models

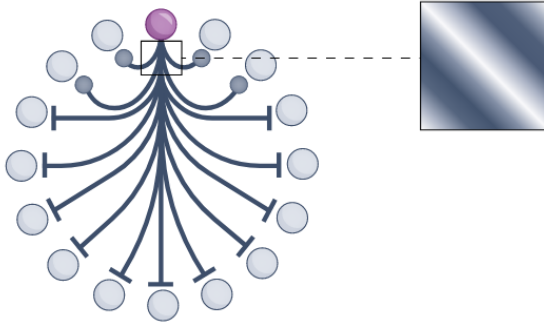
To assess the framework’s merits in making biologically relevant predictions, additional constraints can be added to the loss function. In the simplest case, one may constrain neural activities to be positive and bounded by using nonlinearity $\alpha(z) = \frac{\text{ReLU}(z)}{z+1}$, and minimize "wiring cost" by enforcing a L2 weight decay term. For positivity within the input populations multiplicatively gated by actions, we additionally constrain each actions channel to be positive by injecting positive and negative actions to separate partitions of the network, much like the separation between CW and CCW neurons in the PB.

With these constraints, the network is trained on the 1D ring and benchmarked with the fly ring attractor. In the ring attractor model, EB cells are thought to form stable pattern formation by a symmetric weight distribution with local excitation and global inhibition (Figure 2b for a schematic). In the trained network, the recurrent weight matrix W_{rr} among representation neurons qualitatively captures this distribution (Figure 2a left). On the other hand, PB neurons are thought to introduce CW/CCW limit cycles into the network by modulating the representation neurons with an offset in their preferred direction (Figure 2c), which agrees with the asymmetric connection pattern from input subnetworks of each sign to the representation network (Figure 2 right).

a



b



c

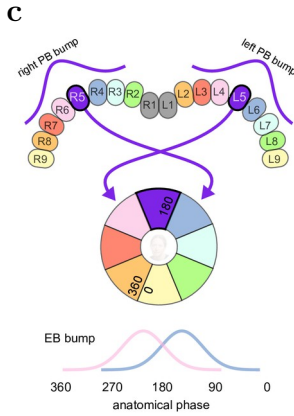


Figure 2 **a** Sorted weight matrices. Representation neurons are sorted by the apparent "angle preference" in their activity patterns, and input neurons are sorted by the representation neuron that most strongly modulates them. Left: Sorted weight matrix within the representation population. Right: Sorted weight matrix from input populations to the representation population, respectively for the CW and CCW population. **b**, **c** Schematics for the symmetric connections within the compass population and the asymmetric connections from two input populations to the compass population, adapted respectively from [2, 4]

4 Conclusion and Plans for Further Work

In conclusion, the first half of the present project carried out in Michaelmas has established a minimal method to discover neural circuits that represent continuous state spaces as attractor manifolds. This method is agnostic to the manifold parametrization for no explicit supervision on manifold positions. Instead, one only assumes that there exists a way to label states on the manifold as similar or different, and the network is trained to respect these labels by contrastive learning on integrating action sequences. In Section 3.1, it is shown that with additional constraints, the network can be trained to recover known neural circuits, such as the ring attractor model.

In steps ahead, the project will investigate how the brain learns to construct and attach meanings to these representations in the motor control context. Specifically, one may interpret manifold elements as transformations, such as rotations, to define a neural population $\mathcal{P}_g$ that hosts abstract behavioral goals, and constrain the null transformation to correspond to a baseline activity profile. Based on goal representations in $\mathcal{P}_g$, a controller K_g would be trained to control a downstream motor population $\mathcal{P}_m$ that serves as a dynamical motor effector (e.g., motor cortex) to perform the desired transformation. We are interested in designing a self-supervision objective that allows $\mathcal{P}_g$ to be learned in conjunction with K_g , during random exploration of the action space, based on self-consistency objectives on motor efference copies and sensory feedback I^* .

Practically, this extension involves exploring potential ways to incorporate the two sources of information. For example, we will experiment with incorporating an additional forward model that makes sensory predictions $\hat{I}$ based on $\mathcal{P}_g$ representations, in turn updating $\mathcal{P}_g$ based on the inferred manifold action from sensory error $I^* - \hat{I}$. This, in conjunction to integration of the inferred action from motor efference, sets the stage for a neural observer-controller system.

Potential difficulties include various types of collapsing solutions and local minima solutions that may arise from the self-supervision objective. For example, the goal population may learn to "cheat" by producing spurious goal representations that satisfy control objectives in collapsing manners. These will be most salient in end-to-end training, where the all components of the control system are trained simultaneously. We plan to address these issues by designing the objective in an incremental manner, where we start with pre-trained parts of the control system and simplified environments, and gradually increase the complexity of the system and the environment. As a potential extension, we may consider a hierarchical control system that mirrors the spirit of the mammalian motor system, where we replace the plant $\mathcal{P}_m$ with another control system, driven by both the high-level control system $\{\mathcal{P}, K\}_g$ and low-level sensory feedback (e.g., proprioception).

References

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